



UNIVERSITÀ DEGLI STUDI DI MILANO

DATA INTEGRATION WITH MOFA ASSIGNMENT 2

Course: Scientific Visualization

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Code and Visualizations available at

<https://github.com/Filippo-Corti/ScientificVisualization/tree/main/02-data-integration>



HPO-GO Dataset Overview

4 Binary Matrices:

- **GO (Gene Ontology):** describes each Gene based on:
 - **CC** (Cellular Components): "Where" the Gene is.
 - **BP** (Biological Processes): "What" the Gene does (at a high level).
 - **MF** (Molecular Functions): "What" the Gene does (at a low level).
- **HPO (Human Phenotype Ontology):** describes Phenotypic Abnormalities:
 - What Symptoms does a Mutation in this Gene have?

For each matrix:

- **rows** = Genes (5183, the same for each view)
- **columns** = ontology terms (24794 across the 4 views)
- **entries** = 0/1 (not annotated/annotated)



Our Goals

Our goals are:

- Experiment with **Data Integration**, understanding:
 - In which way a **Model-Based Approach** is more robust than an **Early Integration**.
 - How to Evaluate the result of a Model-Based Approach (**MOFA**).
- Use the Integrated Data to:
 - Visualize the Dataset using **Data Integration-Specific Visualizations**.
 - Obtain a better **Understanding** of the Dataset through Visualization.

MOFA – Multi-Omics Factor Analysis

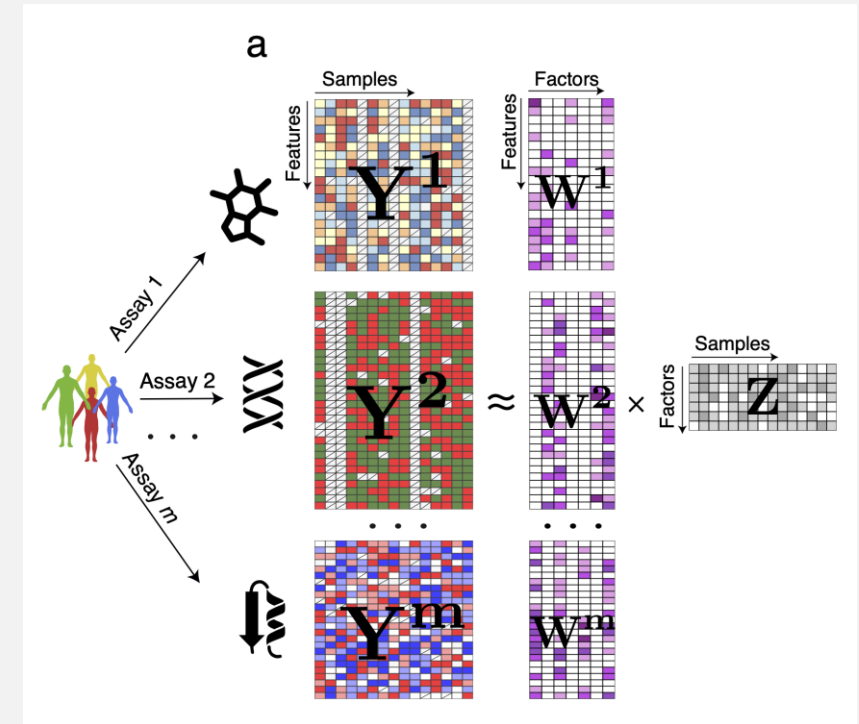
MOFA constitutes a **Model-Based Approach** to Data Integration:

- **Inputs:**
 - Multiple **-Omic Data** Types on the same Set of Samples (or on Overlapping Subsets).
- **Outputs:**
 - An **Interpretable Low-Dimensional Representation** of the Inputs.

It does so by identifying **Sources of Variation**, specific to one View or shared cross multiple ones.

In our Dataset:

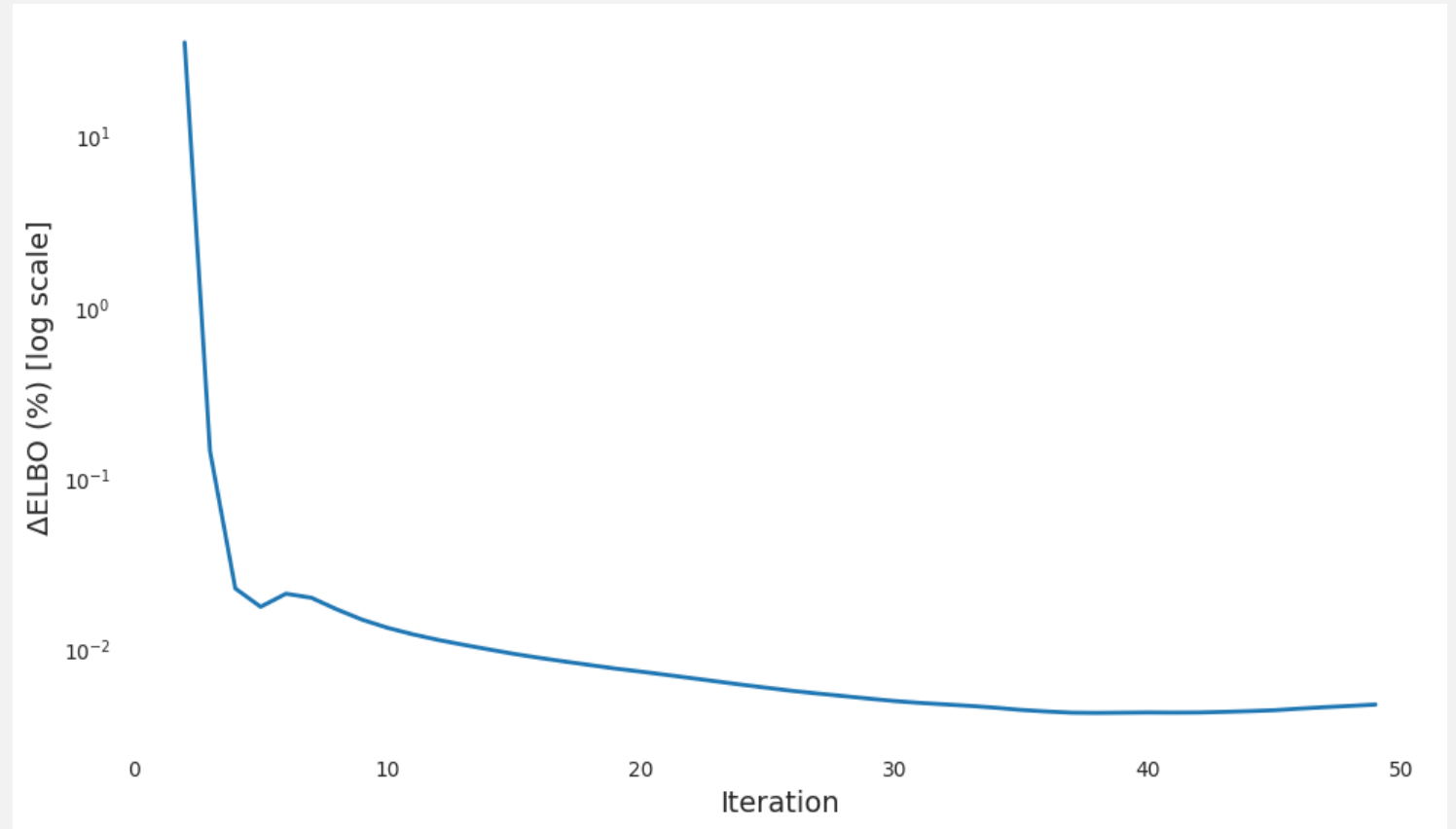
- It captures a **Global Structure** across GO and HPO Annotations.
- It produces a single **Latent Factor Matrix** of Size: $N_{\text{Factors}} \times N_{\text{Samples}}$



MOFA – Training

We ran **MOFA** with the following parameters:

- 4 Views (5183 Samples):
 - **GO-BP: 9873** Features
 - **GO-CC: 1478** Features
 - **GO-MF: 3258** Features
 - **HPO: 10185** Features.
- **Bernoulli Likelihoods**
- **50** Iterations
- **15** Factors

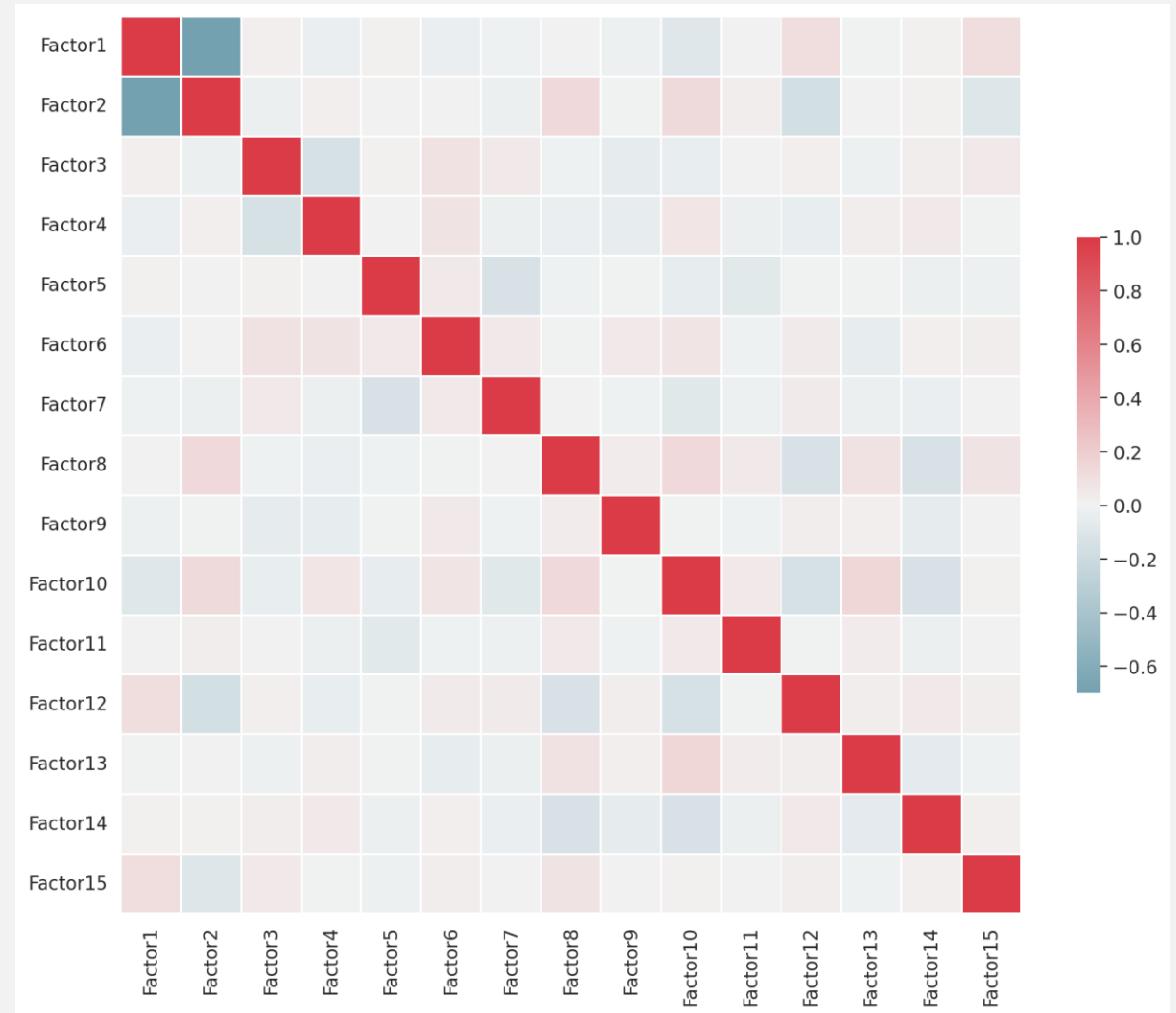


MOFA – Number of Factors

How do we choose 15 Factors?

- The more Factors there are, the less **Interpretable** the Model becomes.
- Computing **Correlation** between the Factors can suggest that certain Factors are **copies/mirrors** of other Factors:
 - In our case, **Factor 1** and **Factor 2** appear highly **inversely correlated**.

Maybe we only need 1?



MOFA – Clustering

We Clustered the Samples (Genes):

- Using the Model from **MOFA**
 - **KMeans** for $k = 2..100$
- On each of the Views (GO-BP, GO-CC, GO-MF, HPO):
 - **MiniBatchKMeans** for $k = 2..100$
 - This is not what we would expect, considering Binary Data.
 - However, KModes would run way too slow.
 - And KMeans is actually considered to be approximating KModes fairly well for high-dimensionality Binary Data.

Result: 495 ways of Clustering the Samples

MOFA – Clustering & NMI

Two Clustering Assignments can be compared using Normalized Mutual Information:

$$\text{MI}(U, V) = \sum_{i=1}^{|U|} \sum_{j=1}^{|V|} P(i, j) \log \left(\frac{P(i, j)}{P(i)P'(j)} \right)$$

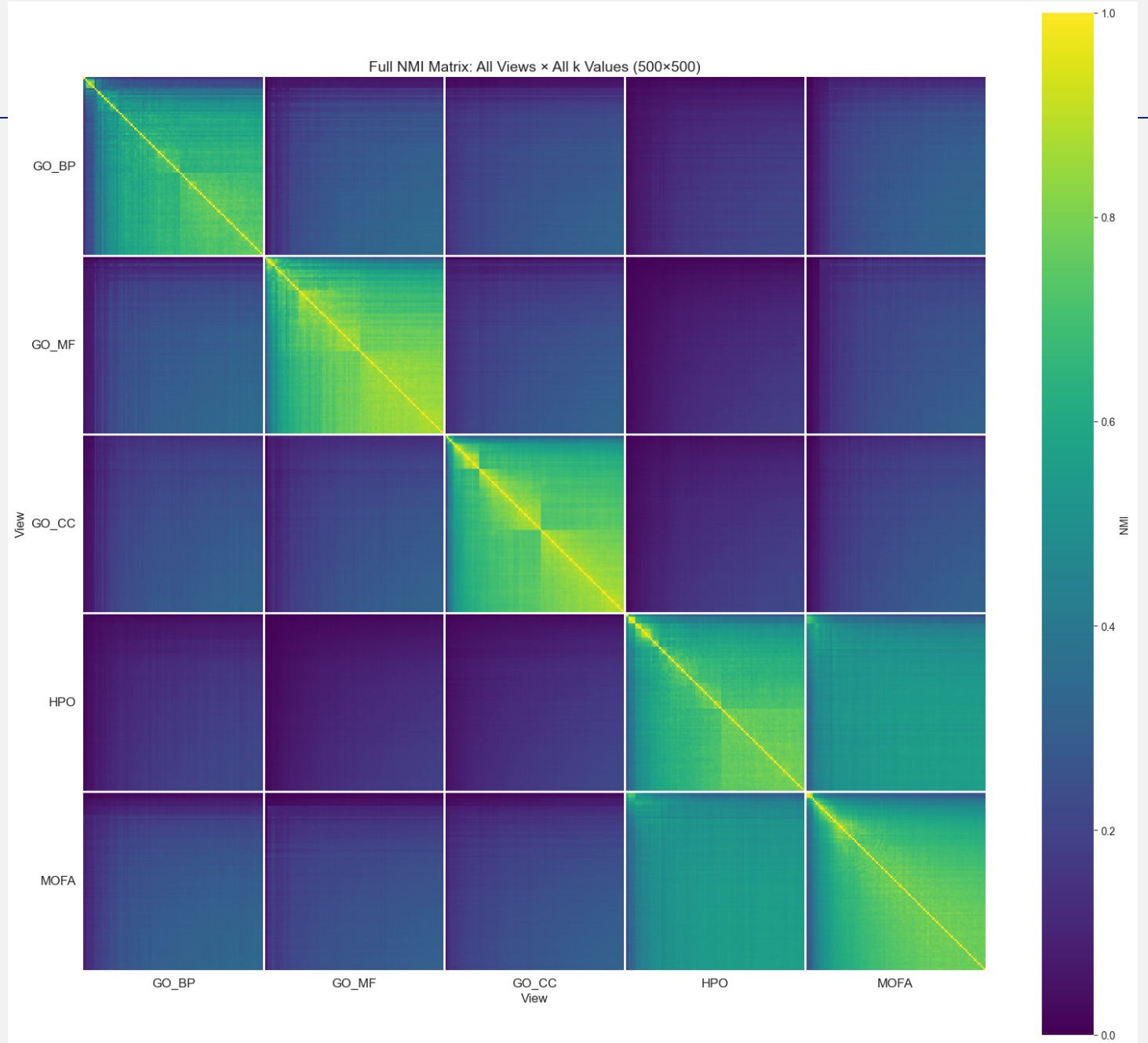
$$\text{NMI}(U, V) = \frac{\text{MI}(U, V)}{\text{mean}(H(U), H(V))}$$

Intuitively, it measures how much one Cluster tells you about the other.
In our case, we can compute a **495x495 NMI Matrix**.

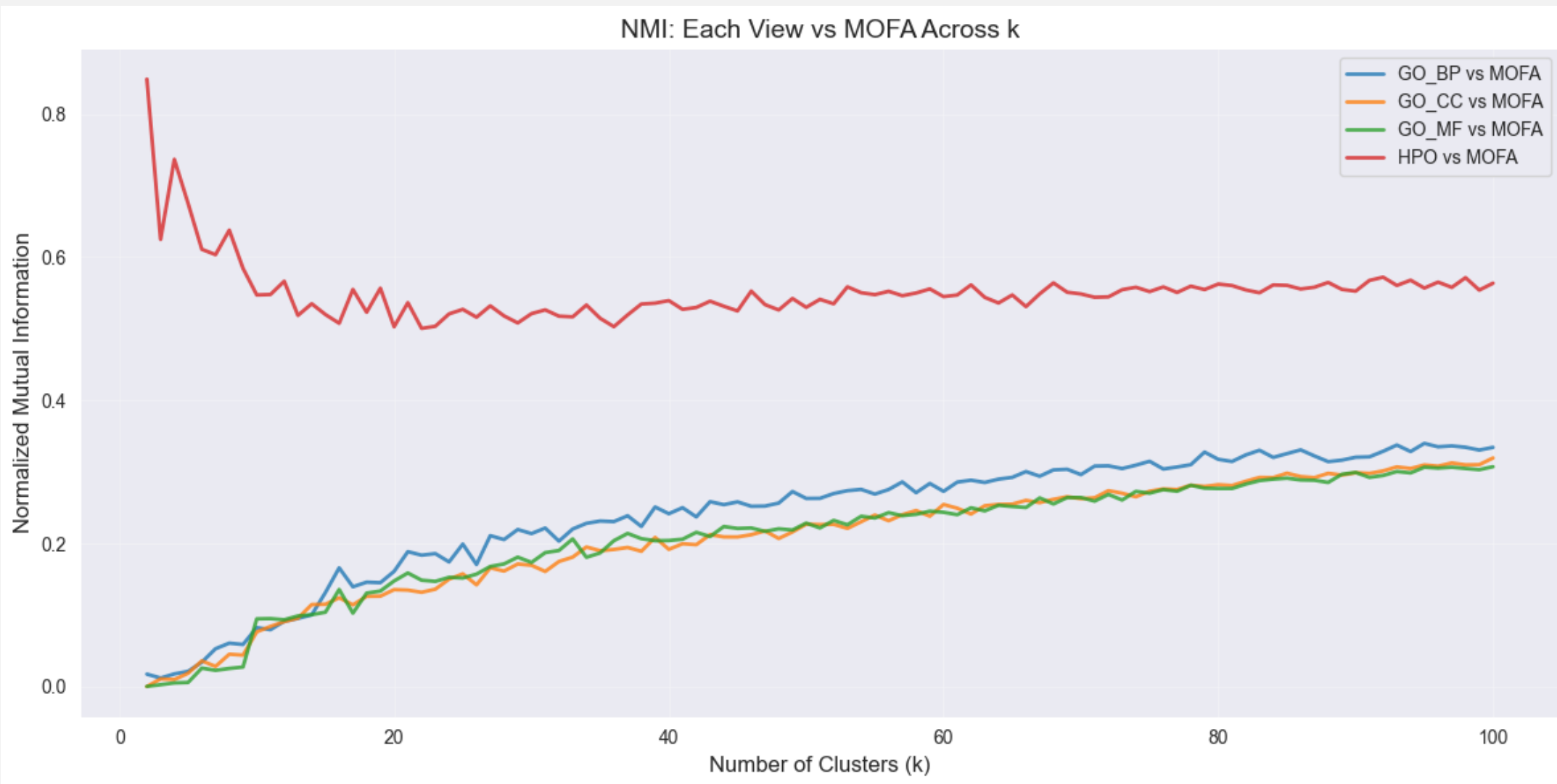
MOFA – NMI

Some observations:

- Each **View-Pair Comparison** appears like a Color Gradient.
- Each View tends to **agree** more with **itself**.
- **HPO** and **MOFA** are unusually **similar**.



MOFA – NMI over different k's

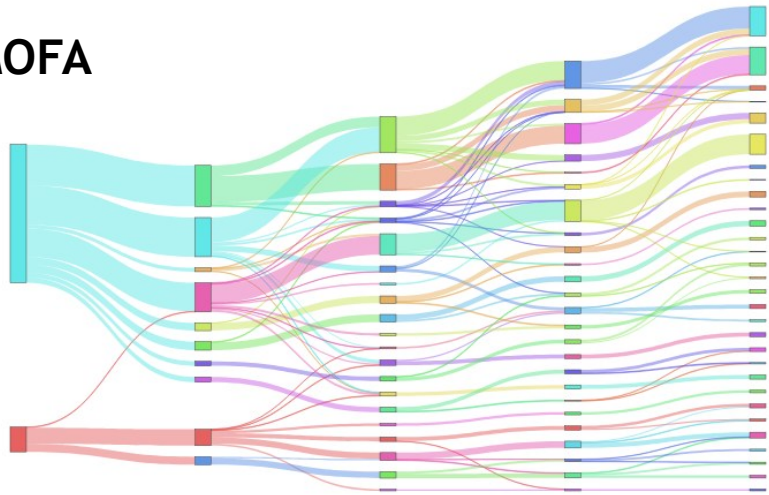


MOFA – NMI Stability over different k's

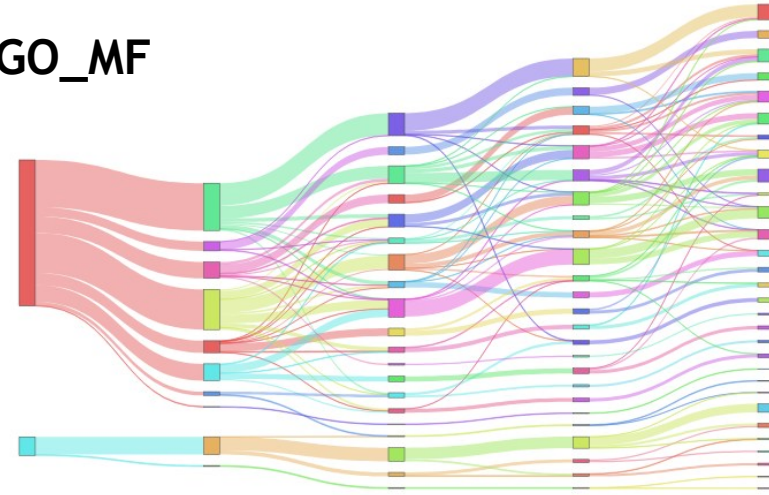


Sankey Plots – $k=2, 10, 20, 25, 30$

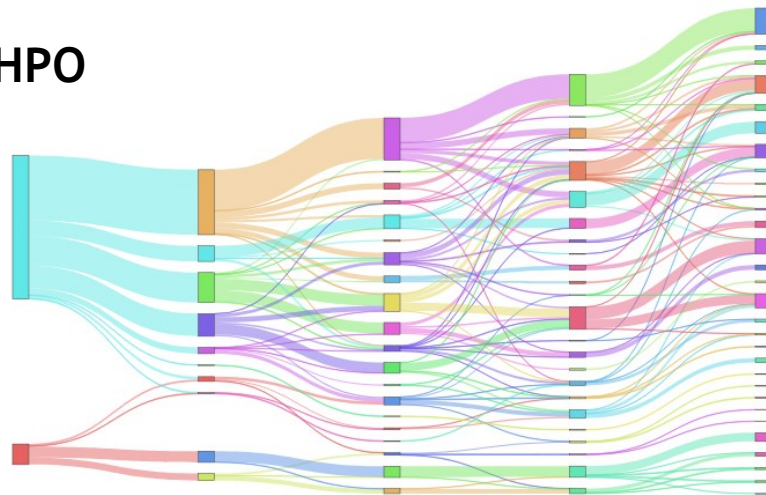
MOFA



GO_MF



HPO

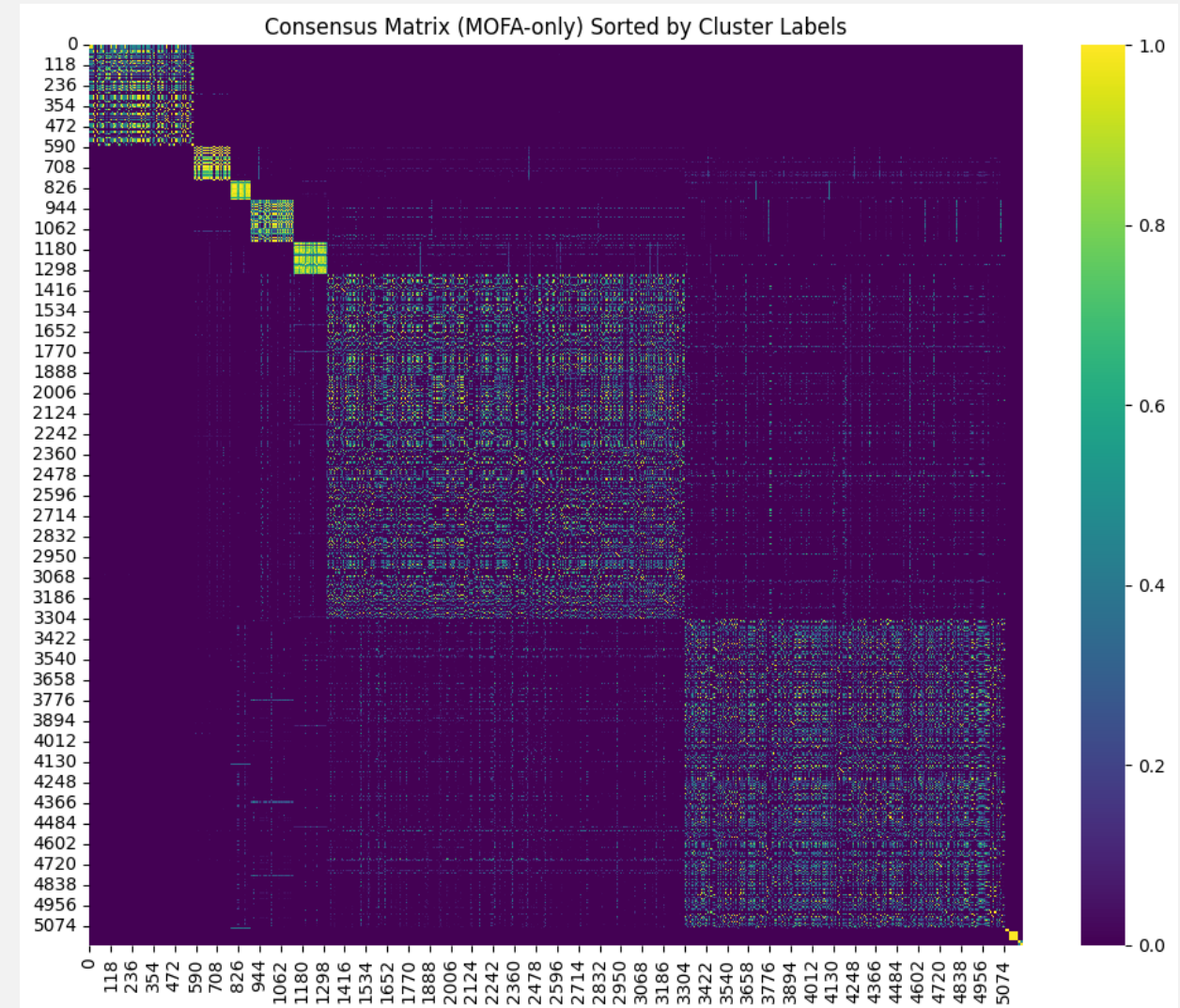


Notes

1. We computed the **consensus matrix** (5138×5138).
 - It represents, for each pair of samples, how often they end up in the same cluster.
2. We computed the **CDF** (Cumulative Distribution Function) of this matrix:
 - for a given consensus value (e.g., 0.8), it tells how many sample pairs have a value $\leq$ that threshold.
3. We computed the **Area Under the CDF** (AUC), which itself behaves like a cumulative stability curve.
4. For each value of k in the interpretable range (2-20), the AUC provides an overall measure of **clustering stability**.
5. We computed the Δ CDF and based on the stability jump we selected $k = 10$.
 - Each cluster is composed of a group of samples.
6. Using **G.Profiler**, we extracted the columns/features that typically characterize the samples in each cluster.
 - For each term (BP, CC, MF), we also obtain a **descriptive explanation**.
7. This gives us a hierarchy of grouped columns, aligned with the structure of the $k = 10$ clusters.
 - Each enriched term comes with its textual description.
8. At this point, **we can plot the graph**.

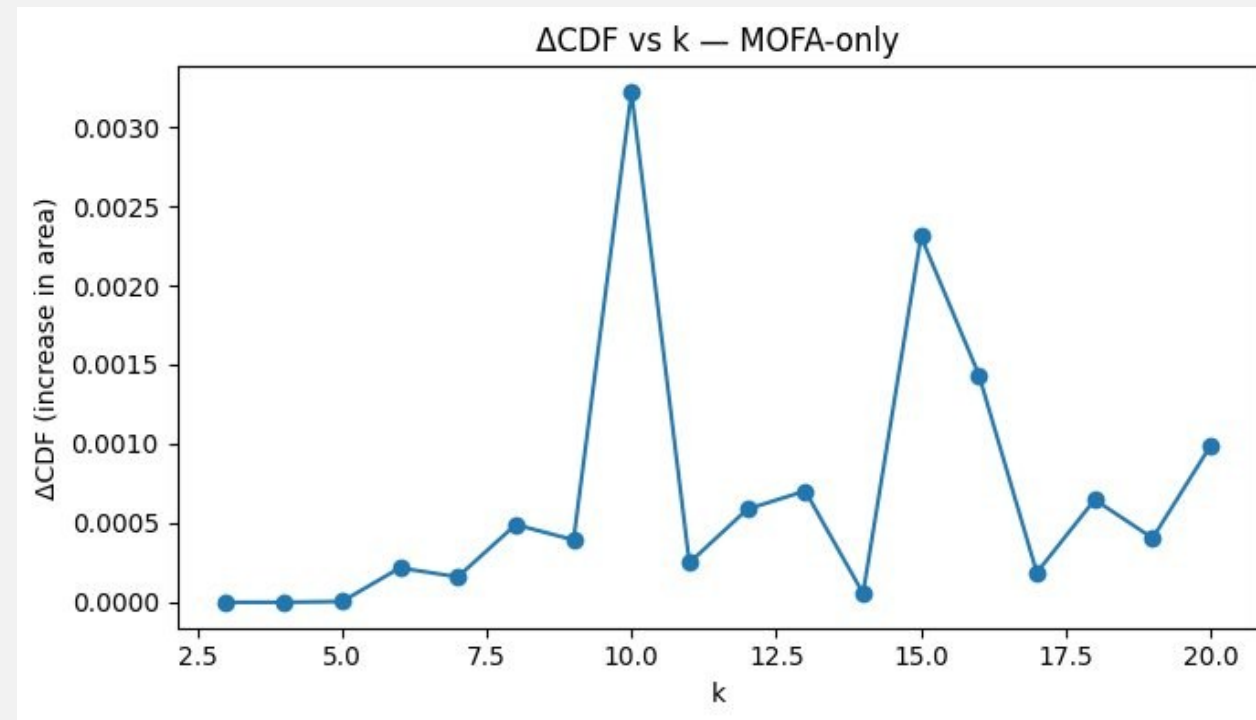
Consensus Clustering

- Consensus clustering asks: across all clustering results I have, **which genes repeatedly end up together?**
- It combines results from **multiple clustering runs** and identifies **stable and reproducible clusters**.
- Consensus clustering builds a **consensus matrix**, which is a sample-by-sample matrix. Each entry (i, j) means:
 - **1.0** → samples always end up in the same cluster across repeated clustering cycle.
 - **0.0** → samples never cluster together.



Consensus Clustering – find optimal k

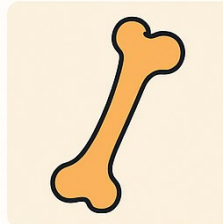
- Build the **CDF curve**: $CDF(x)$ = proportion of sample pairs with consensus $\leq x$
- Calculate the **Area Under the CDF (AUC)**
- Compute $\Delta CDF(k)$:
 - **Large ΔCDF** → adding clusters improves stability.
 - **Small ΔCDF** → additional clusters not meaningful.
- We selected $k = 10$



Functional Annotation of Consensus Clusters

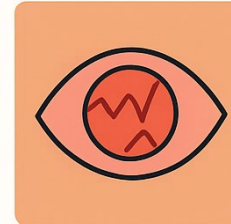
- After obtaining **consensus gene clusters**, we assign **biological meaning** using functional enrichment.
- ***g:Profiler*** tests whether each cluster's genes are enriched for known biological categories.
- For each cluster:
 - Input: genes in the cluster
 - Background: all measured genes
 - Output: significant functional terms
- From the enrichment table:
 - Keep **GO:BP** entries
 - Select the **top N most significant terms**

Cluster 1



Abnormal long bone & limb morphology / skeletal development

Cluster 2



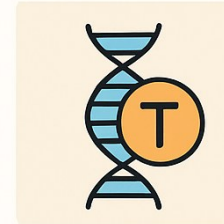
Retinal degeneration / visual electrophysiology defects

Cluster 3



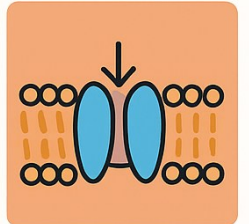
Epileptic EEG abnormalities / neurophysiology

Cluster 4



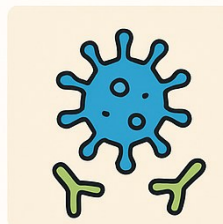
Transcription regulation & DNA-binding transcription factors

Cluster 5



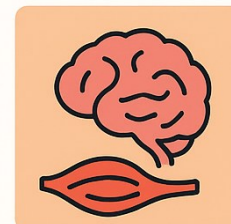
Ion transport transmembrane transport

Cluster 6



Immune abnormalities + extracellular region components

Cluster 7



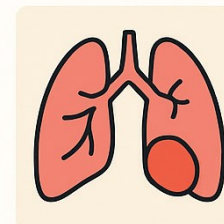
Neurodevelopmental + muscular physiology abnormalities

Cluster 8



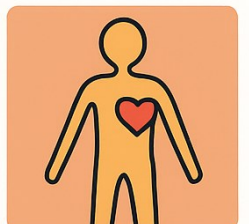
Mixed neuro-ocular-vascular syndrome features

Cluster 9



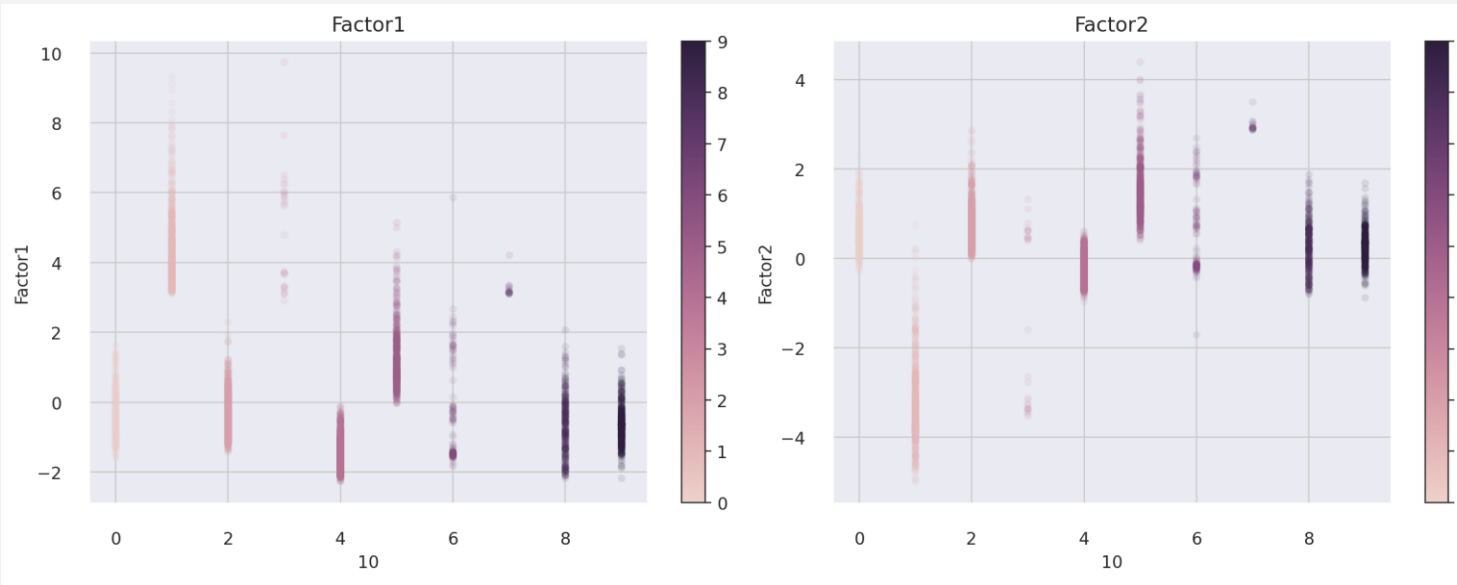
Pulmonary, vascular situs abnormalities+ reproductive

Cluster 10

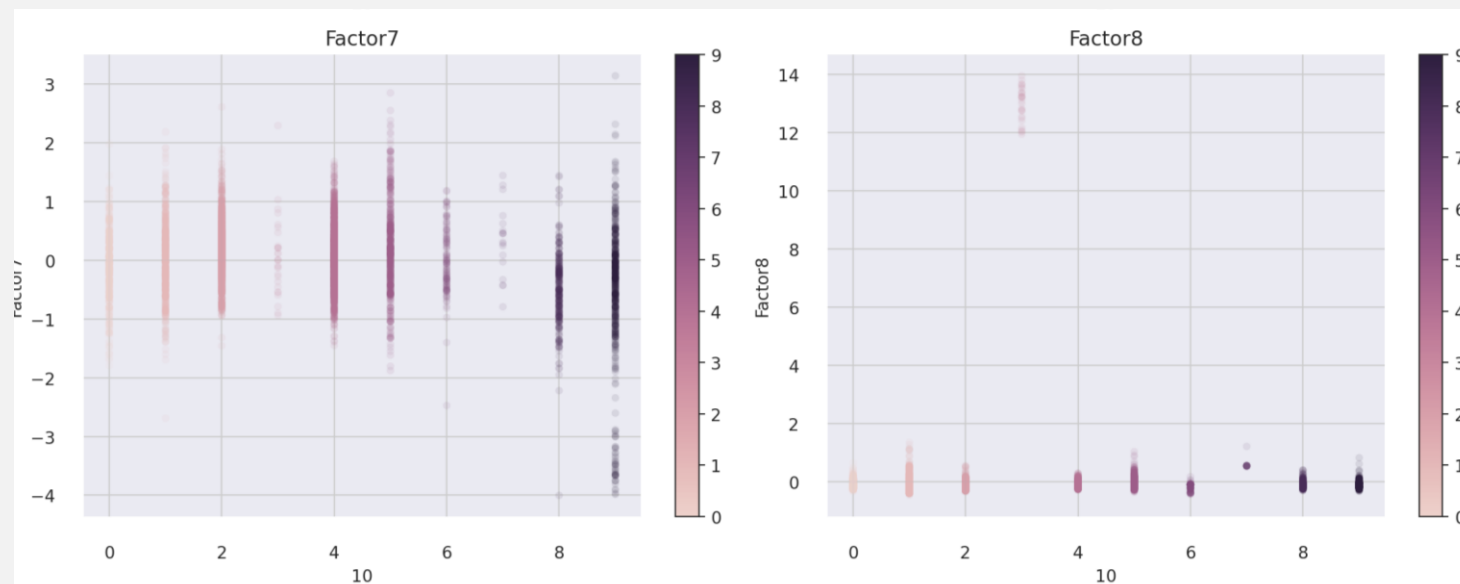


Marfanoid skeletal & vascular abnormalities

MOFA – Factors by Cluster



- **Factor 1 and Factor 2** appear highly inversely correlated.

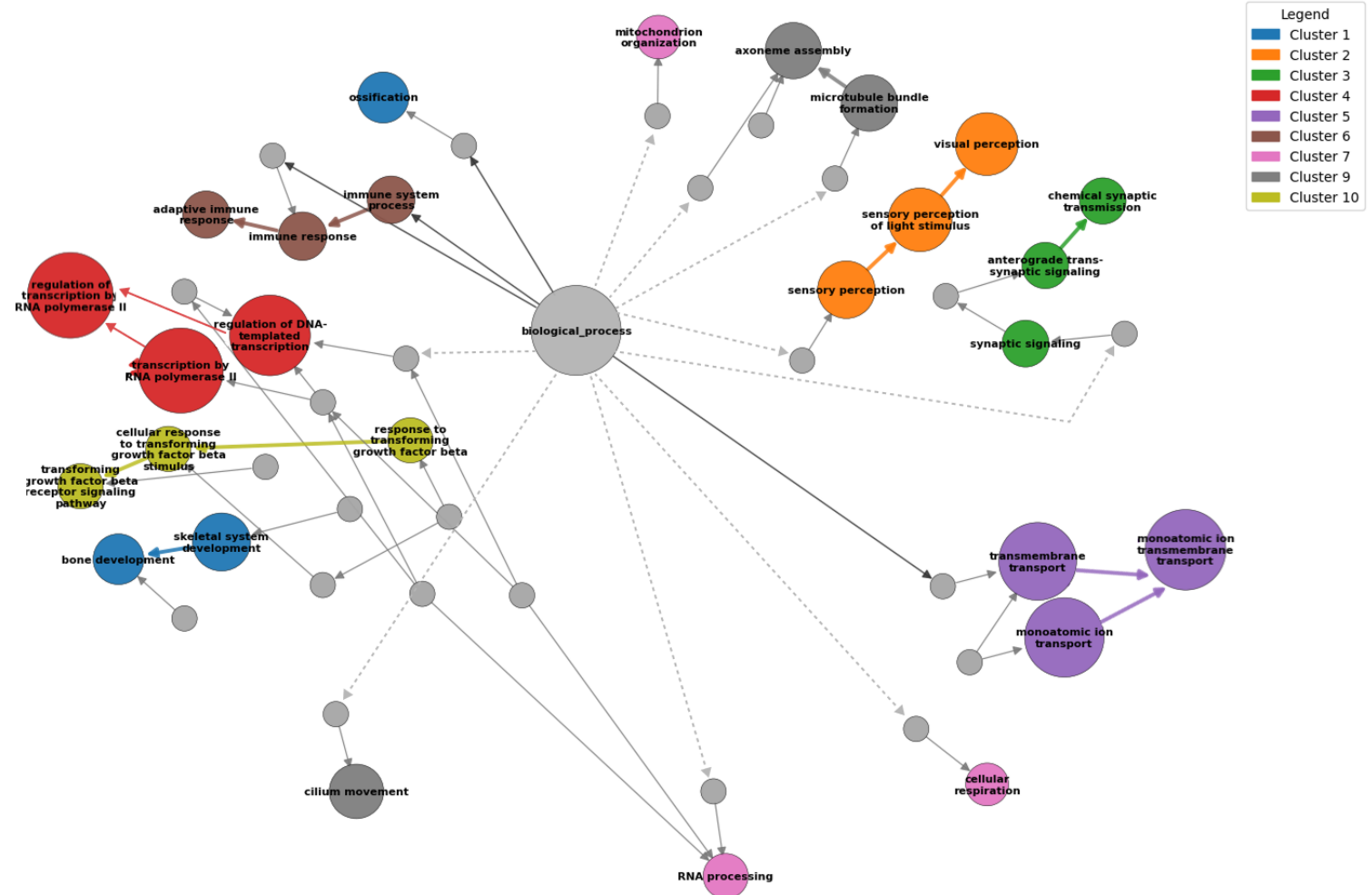


- The genes in **cluster 3** share biological processes strongly represented by **Factor 8**.
- This cluster is likely enriched for a **specific** biological pathway.

GO Biological Process Graph

Start from **top enriched GO:BP terms per cluster**:

- **Node color:** indicates cluster membership.
- **Edges:** represent GO hierarchy.
- **Node size:** reflects how strongly enriched that GO term is in the cluster.





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Thank you for your attention.

The MOFA Framework:
<https://biofam.github.io/MOFA2/>

G:Profiler Service:
<https://biit.cs.ut.ee/gprofiler/gost>

Understanding of the Biomedical Information about
Genes and their Ontology was achieved with the
help of ChatGPT

